

1. What does “mobile” actually mean?

A trap people fall into: they think “mobile” = phone / iPad / tablet.

Your slide says:

Fundamentally, ‘**mobile**’ refers to the **user**, not the device or the application.

So:

- A laptop can be “mobile” if the **user** is moving (train, café).
- A handheld scanner, a Gameboy, a delivery driver’s pad, an iPod → all can be part of “mobile” experience.

MEMORIZE:

“Mobile refers to the *user*, not the device or the application.”

Because exam MCQ can ask “what is mobile anyway?”

2. The Carry Principle & PCD (Personal Communication Device)

Your slides talk about the **Carry Principle** using “PCD”.

A **PCD (Personal Communication Device)** has four main properties:

1. **Personal**

- Belongs to one person.
- Personally identifiable (SIM, phone number, account).
- Has a messaging address and ongoing service.

2. **Communicative**

- Can send/receive **messages** in various forms: calls, SMS, push notifications, data.

3. Handheld

- **Portable**, small enough to carry.
- Can be used with **one hand** (even if two hands / a surface are more convenient in practice).

4. Wakable

- Can wake quickly, either:
 - By **user** (press button, tap screen), or
 - By **network** (incoming call, notification, alarm).

👉 **MEMORIZE: 4 PCD properties = Personal, Communicative, Handheld, Wakable.**

2.1 Carry Principle implications

Your slides then say “Carry Principle implications” for: **Form, Features, Device Capabilities, User Interface, Proliferation.**

In simple words:

- **Form & Features**
 - Device must be **small**, light, easy to carry.
 - Hardware is optimized for being on you all the time (pocket, bag).
- **Device capabilities**
 - Because we “carry” them, they tend to get:
 - Sensors (GPS, camera, mic, accelerometer...)
 - Multiple communication options (Wi-Fi, data, Bluetooth).
- **User Interface – Single window operation**

- You have limited screen, often **one window at a time**.
- No giant multi-window desktop like a PC.
- So UI must be **focused and simple**, avoid clutter.
- **Proliferation**
 - Because devices are easy to carry, they become **very common** (everyone has one), and people start using them for many things: money, social, work.

2.2 Carry Principle for PCD Users

This is about how **users behave** when they carry these devices all the time.

- **User availability**
 - People are reachable almost **all the time** (calls, WhatsApp, NayaPay alerts).
- **Sustained focus (or lack of)**
 - They **don't always give full attention** to the app.
 - They might be walking, in a meeting, talking to someone.
- **Social behavior**
 - Phone use is part of social life: checking messages at dinner, sending screenshots, sharing payment screenshots.

So when designing NayaPay / any app, you must assume:

User is **busy, distracted, and social**, not sitting calmly with full focus like on a big desktop screen.

3. Components of a Mobile App

Your slides give a nice breakdown of what makes up a **mobile application ecosystem**.

3.1 Components (1/2)

1. A PCD

- With its own use metaphor (e.g., “springboard” on iOS), browser, app environment, and capabilities.

2. A user

- Could be:
 - On a train
 - In a meeting
 - In a restaurant
 - Walking down the street
- Might be:
 - Focused fully on the app
 - Or half-attending a conversation.

3. One or more application platforms

- Web browsers, native app environments (iOS, Android), messaging environments (SMS, WhatsApp), media environments.
- New capabilities appear regularly.

4. One or more output interfaces to the outside world

- Screen
- Speaker
- Infrared
- Bluetooth
- Local wireless (Wi-Fi)

- Cellular wireless
- Unique terminal identification (IMEI, etc.)

3.2 Components (2/2)

5. One or more input interfaces

- Keypad (limited)
- Touchscreen
- Microphone
- Camera
- RFID reader
- GPS
- Infrared, Bluetooth, Wi-Fi, cellular, etc.

6. Optional server infrastructure

- Backend servers that:
 - Store data
 - Process transactions
 - Connect to external APIs/services.

7. A network and carrier

- Wireless operator that:
 - Connects user to Internet/other users
 - Sells data/services
 - Sometimes restricts which devices are allowed or what can be done on the network.

👉 For exam: **MEMORIZE at least 4 components** (PCD, user, platforms, input/output interfaces, server, network).

4. Characteristics of a Mobile User

Your slide lists these characteristics:

- **Mobile**
- **Interruptible and easily distracted**
- **Available**
- **Sociable**
- **Contextual**
- **Identifiable**

Let's break them.

1. **Mobile**

- Always on the move; context changes: sitting, commuting, walking.

2. **Interruptible & easily distracted**

- Notifications, real-world interruptions, calls, people talking → you cannot assume continuous focus.

3. **Available**

- Because they carry their phone, you can reach them almost any time (push notifications, calls).

4. **Sociable**

- Phones are tools for **social interaction** (chat, video, sending screenshots, sending money).

5. Contextual

- The context (time, location, activity) matters a lot. More below.

6. Identifiable

- Device is tied to a person (SIM, login, biometrics), so apps can identify the user.

👉 **MEMORIZE** this list of 6 words.

4.1 Mobile apps as contextual

Your slide gives examples of using **context** smartly:

1. Calendar app example

- App could:
 - Switch ringer to **vibrate** during a meeting.
 - Tell the caller “user is in a meeting now”.
 - Let caller mark **urgency**.
 - Let recipient decide whether to accept.

2. Travel companion app

- Uses:
 - Location
 - Flight number
 - Flight status
 - Traffic conditions
- Alerts user when they need to **leave for airport**.

- Notifies meeting attendees if user is going to be late.

3. Restaurant coupon app

- Sends coupons at **lunchtime**, when user is **away from home** and **near restaurants**.

Future devices:

- Could have accelerometers, temperature sensors, fingerprint readers, etc.

Key idea:

Mobile UX should be **context-aware**: time, place, activity, social situation.

5. Groups and tribes – social angle of mobile

Your slide: “Groups and Tribes” → phones are part of social identity.

- **Voice and texting**: basic communication.
- **Extending online communities**: mobile apps for Discord, Reddit, WhatsApp groups.
- **Physical and mobile hybrids**: location-based meetups, Pokémon Go style uses.
- **Mobiles as status**: phone model / case / AirPods as **status symbol**.

For HCI: this means that:

- Design can foster **community** (e.g., NayaPay P2P features, gift envelopes).
 - Design can signal **status** (premium animations, card design, etc.).
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6. Types of mobile devices by use

Your slide: **Use of Mobile Device** – 4 categories.

1. **General-purpose work**

- Multi-purpose devices near user **while at work**.
- Example: company-issued smartphone.

2. **General-purpose entertainment**

- Multi-purpose devices with **entertainment focus**.
- Used when fun is acceptable – gaming phone, Netflix tablet.

3. **General-purpose communications and control**

- Personal device used for:
 - Voice & text communication
 - Controlling things: smart home, banking, finances.

4. **Targeted devices**

- Devices meant for 1 or few tasks.
- Form varies with purpose (card readers, scanners, POS terminals).

NayaPay would mostly sit under **general-purpose communications and control**: personal phone that handles money and communication.

7. iOS / Apple HIG – Design Themes

Now the Apple part. You have two sources:

- From the mobile design slides: **Deference, Clarity, Depth**.
- From the Week 11 sheet: full **Apple HIG list of 8 principles**.

Let's first unify the principles.

7.1 Apple HIG core principles (Week 11 sheet)

According to your Week 11 activity handout, Apple HIG includes:

1. **Clarity**
2. **Deference**
3. **Depth**
4. **Feedback & Responsiveness**
5. **Direct Manipulation**
6. **Accessibility / Adaptability**
7. **Aesthetic Integrity**
8. **Use of Standard Components**

👉 **MEMORIZE** all 8 names exactly.

7.1.1 Clarity

UI should be easy to understand: clear language, icons, layouts.

- Text readable at all sizes.
- Icons precise and meaningful.
- Layouts not confusing.

Example (NayaPay):

Labels like “Send Money”, “Pay Bill”, “Add Money” must be **obvious**. Don’t write “Operations” or vague words.

7.1.2 Deference

Interface should **defer to the content**; content is the star, UI is the support.

- No heavy borders, gradients, decoration that distracts.
- Background is neutral; content (transactions, balances) stands out.

Example:

On a transaction detail screen, you show:

- Amount, recipient, date, status clearly.
 - The screen chrome (top bar, buttons) is minimal.
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7.1.3 Depth

Use visual layers, shadows, blur, motion to show **hierarchy and spatial relationships**.

Your other slides show examples in Calendar: year → month → day transitions.

- When you tap a month, year view **zooms in**.
- When you tap a day, month splits to show detailed day view.
- Each transition reinforces “where you are” in hierarchy.

Example (NayaPay):

Home → Transactions list → Transaction details:

- List slides left, detail slides in on top → user feels they “went deeper”.
 - Back button shows the previous level.
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7.1.4 Feedback & Responsiveness

UI should respond **immediately** with meaningful feedback: animations, haptics, loading indicators.

- Button tap → highlight or bounce.

- Long process → spinner or progress bar.
 - Successful payment → confirmation screen, tick icon, maybe vibration.
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7.1.5 Direct Manipulation

Use familiar iOS gestures (swipe, drag, pinch); users should **directly manipulate UI elements**.

- Swipe to delete a transaction.
 - Drag slider to choose amount.
 - Pull to refresh transaction list.
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7.1.6 Accessibility / Adaptability

Support Dynamic Type, proper touch target size, VoiceOver, color contrast.

- Font size adjustable.
 - Buttons not tiny (at least ~44x44 points).
 - High contrast for text.
 - VoiceOver labels for screen-reader users.
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7.1.7 Aesthetic Integrity

Visual design should reflect app's **purpose and behavior**; simple, consistent, meaningful.

- Banking app: calm, serious, trustworthy colors (not neon clown palette).
- Layout consistent across screens.

- No random style change on one screen.
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7.1.8 Use of Standard Components

Use **native iOS UI components** when possible for consistency and familiarity.

- Use standard navigation bars, tab bars, table views.
- Use system alerts, system date pickers if possible.

This matches your Week 11 class tasks like “what standard iOS components would you like to use in NayaPay Help/Support”.

7.2 iOS Design Steps (from the slides)

Your mobile design slides also say this:

Step 1:

- Look past the UI to core functionality.
- Use iOS themes (deference, clarity, depth) to shape UI.
- Add details carefully, never gratuitously.
- Design UI to adapt to various devices/modes.
- Be prepared to **defy precedent**, question assumptions, focus on content & functionality.

Step 2:

- Take advantage of the **full screen**.
- Reconsider skeuomorphic realism.
- Use **translucent** elements to hint at content behind.

- Use plenty of **negative (white) space**.
- Let **color simplify** the UI.
- Use **system font** for legibility.
- **Borderless buttons**.

Examples given:

- Folders with translucent backgrounds “floating” above Home.
- Reminders stacking lists in layers.
- Calendar transitions for depth.

👉 **MEMORIZE keyword-style:**

- Step 1 = focus on **core functionality**, use themes, adapt, question assumptions.
- Step 2 = **full screen, negative space, translucency, system font, borderless buttons, depth transitions**.

8. Week 11 Class Activity – How it connects

Your activity sheet gives 9 tasks inside NayaPay and always asks:

- What strong/weak points based on HIG?
- What gestures?
- What feedback?
- What deference?
- What standard components?

The **pattern** your teacher wants:

- **Look at a flow** (add money, block card, check history, update phone number, pay bill, contact support, send gift, pay challan).
- For each step, apply **HIG principles**:
 - Is UI clear?
 - Is content the hero (deference)?
 - Is there depth/hierarchy visible?
 - Is there immediate feedback?
 - Are we using gestures (direct manipulation)?
 - Is it accessible?
 - Is aesthetic integrity right for a finance app?
 - Are we using standard components?

This is exactly the kind of case-study question you might get.

PRACTICE SECTION

Now: **5 MCQs, 5 Short Qs, 1 Case Study** – all answered and explained.

A. 5 MCQs

MCQ 1

In the context of your Week 11 slides, what does the term “**mobile**” fundamentally refer to?

- A. Any device that has a SIM card
- B. Any app that runs on a smartphone

- C. The user, who may be moving and using different devices
- D. Devices with Wi-Fi and Bluetooth

Correct answer: C

Explanation:

Your slides explicitly say that fundamentally, “mobile” refers to the **user**, not the device or application; the user may move between contexts and devices.

MCQ 2

Which of the following is **NOT** one of the four defining properties of a **Personal Communication Device (PCD)**?

- A. Personal
- B. Communicative
- C. Foldable
- D. Wakable

Correct answer: C

Explanation:

The PCD is defined as **Personal, Communicative, Handheld, Wakable**. “Foldable” is not one of the properties.

MCQ 3

Which Apple HIG principle focuses on **clear language, icons, and layouts** so that the UI is easy to understand?

- A. Depth
- B. Aesthetic Integrity
- C. Clarity
- D. Deference

Correct answer: C

Explanation:

Clarity in Apple HIG means the UI should be easy to understand, with clear text, icons, and layouts.

MCQ 4

A mobile calendar app that **automatically switches the phone to vibrate during meetings** and informs callers that the recipient is busy is primarily demonstrating which characteristic of mobile users or devices?

- A. Identifiability
- B. Contextual behavior
- C. Aesthetic integrity
- D. Standard components

Correct answer: B

Explanation:

This is an example of a **contextual** mobile experience, where the app uses information like time and calendar events to adapt behavior.

MCQ 5

Which HIG principle is MOST directly involved when a user **swipes left to delete a transaction** or **drags to reorder items** in a list?

- A. Feedback & Responsiveness
- B. Direct Manipulation
- C. Deference
- D. Use of Standard Components

Correct answer: B

Explanation:

Direct Manipulation refers to letting users interact with content using familiar gestures like swipe, drag, and pinch, so they feel like they are directly manipulating the elements.

B. 5 Short Answer Questions

SAQ 1

List the four main properties of a Personal Communication Device (PCD) and explain each briefly.

Model answer:

1. **Personal** – The device generally belongs to one person, is identifiable (e.g., phone number), and has a messaging address and ongoing service.
 2. **Communicative** – It can send and receive messages in various forms, such as voice, SMS, or data.
 3. **Handheld** – It is portable and can be operated with one hand, even if using two hands is sometimes easier.
 4. **Wakable** – It can be quickly awakened by the user or by the network, for example by tapping the screen or receiving a call/notification.
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SAQ 2

What does it mean when we say mobile users are “contextual”? Give one example from the slides.

Model answer:

Mobile users are “contextual” because their tasks and needs depend heavily on their situation (time, place, activity). Apps can use context like location, calendar, or time to adapt behavior. For example, a travel companion app might use the user’s location, flight number, flight status, and traffic conditions to alert them fifteen minutes before they need to leave for the airport.

SAQ 3

State the three main themes Apple emphasizes in iOS design: Deference, Clarity, and Depth. Explain each in one line.

Model answer:

- **Deference** – The interface defers to the content, staying minimal so content remains the focus.
- **Clarity** – Text, icons, and layouts are clear and legible, making the UI easy to understand.
- **Depth** – Visual layers and motion create a sense of hierarchy and space, helping users understand where they are in the app.

SAQ 4

What is meant by “Use of Standard Components” in Apple HIG, and why is it important?

Model answer:

“Use of Standard Components” means using native iOS UI elements (like standard navigation bars, tab bars, table views, alerts) whenever possible. It’s important because these components behave consistently across apps, which makes the interface more familiar and predictable for users, reducing learning effort and mistakes.

SAQ 5

Give two examples of quantitative usability measures you might collect in a mobile usability test, according to your Week 11 materials.

(This connects Week 11 to earlier usability content.)

Model answer:

Two examples are:

1. **Time to complete a task**, such as the time taken to pay a bill in a banking app.
 2. **Number and type of errors per task**, such as how many times users choose the wrong option or enter invalid data while performing a transfer.
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C. Case Study (with full solution/explanation)

Scenario

You are redesigning the **NayaPay mobile app** for iOS.

Your task is to improve the “**Send Money**” flow using mobile design principles and Apple HIG.

Current issues (hypothetical, but very realistic):

- Users often **tap the wrong contact** because the list is dense and text is small.

- After sending money, some users are **unsure if the transaction succeeded**, especially if their phone lags.
- New users struggle to find the **transaction history** to double-check yesterday's transfers.

Questions:

1. Using **mobile user characteristics**, name **two characteristics** that are relevant to this flow and explain why.
2. Using **Apple HIG principles**, propose **three concrete design improvements** for the "Send Money" flow.
3. For each improvement, name the **HIG principle** it reflects (e.g., Clarity, Feedback & Responsiveness, Direct Manipulation, etc.).

Model Answer

1. Relevant mobile user characteristics

Two useful characteristics:

- **Interruptible and easily distracted**
 - Users might be sending money while walking, talking, commuting, or watching TV. They can lose focus quickly, so the flow should be short, clear, and forgiving to errors.
- **Contextual**
 - The importance of the transaction depends on context (e.g., paying a bill on due date vs sending casual money). The app should consider context like time, amount, and recent failures to communicate clearly (extra confirmation for large or urgent transfers).

(You could also mention "Available" or "Identifiable", but these two are the most directly relevant to the problems.)

2. Three design improvements + HIG mapping

Improvement A – Make recipient list clearer and easier to scan

- Increase **font size** for recipient names and amounts using system font.
- Add **profile avatars** or initials for each contact.
- Add more **spacing** (negative space) between contact rows.

HIG principles:

- **Clarity:** Larger text and clear icons make it easier to identify the correct contact.
 - **Accessibility / Adaptability:** Larger touch targets and better legibility help all users, including those with visual difficulties.
 - **Use of Standard Components:** Use standard iOS list/table components for consistency.
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Improvement B – Strong confirmation feedback after sending money

- After user presses “Send”, show:
 - A **full-screen confirmation** with a **large green tick**.
 - Key details: amount, recipient, date/time, reference ID.
 - Optional subtle **haptic feedback** (vibration) on success.
- If network is slow, show a **progress state** (“Processing payment...”) with a spinner.

HIG principles:

- **Feedback & Responsiveness:** User immediately sees that their action is being processed and then sees clear confirmation on success.
- **Deference:** The confirmation screen is simple and focused on transaction details (content), not decorative UI.

- **Depth:** The confirmation can “slide up” from below or come forward visually, indicating a new layer.
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Improvement C – Easier access to transaction history through direct manipulation

- In the “Send Money” screen, allow user to **swipe down** from the top to reveal **recent transfers**.
- Or provide a standard “**History**” **tab** in the bottom tab bar that opens a layered list of past transactions.
- Each transaction row can be **swiped** for options (view details, download receipt).

HIG principles:

- **Direct Manipulation:** Gestures like swipe down or swipe left let users work with content directly (transactions).
 - **Depth:** Recent transactions appear as another layer sliding into view, reinforcing hierarchy (current task vs history).
 - **Use of Standard Components:** Using a standard **tab bar** and list view for transaction history.
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Overall, this answer shows your teacher that you can:

- Connect **mobile characteristics** (distracted, contextual users)
- Apply **Apple HIG principles** (clarity, feedback, direct manipulation, etc.)
- Propose **concrete UI changes** (not just theory).